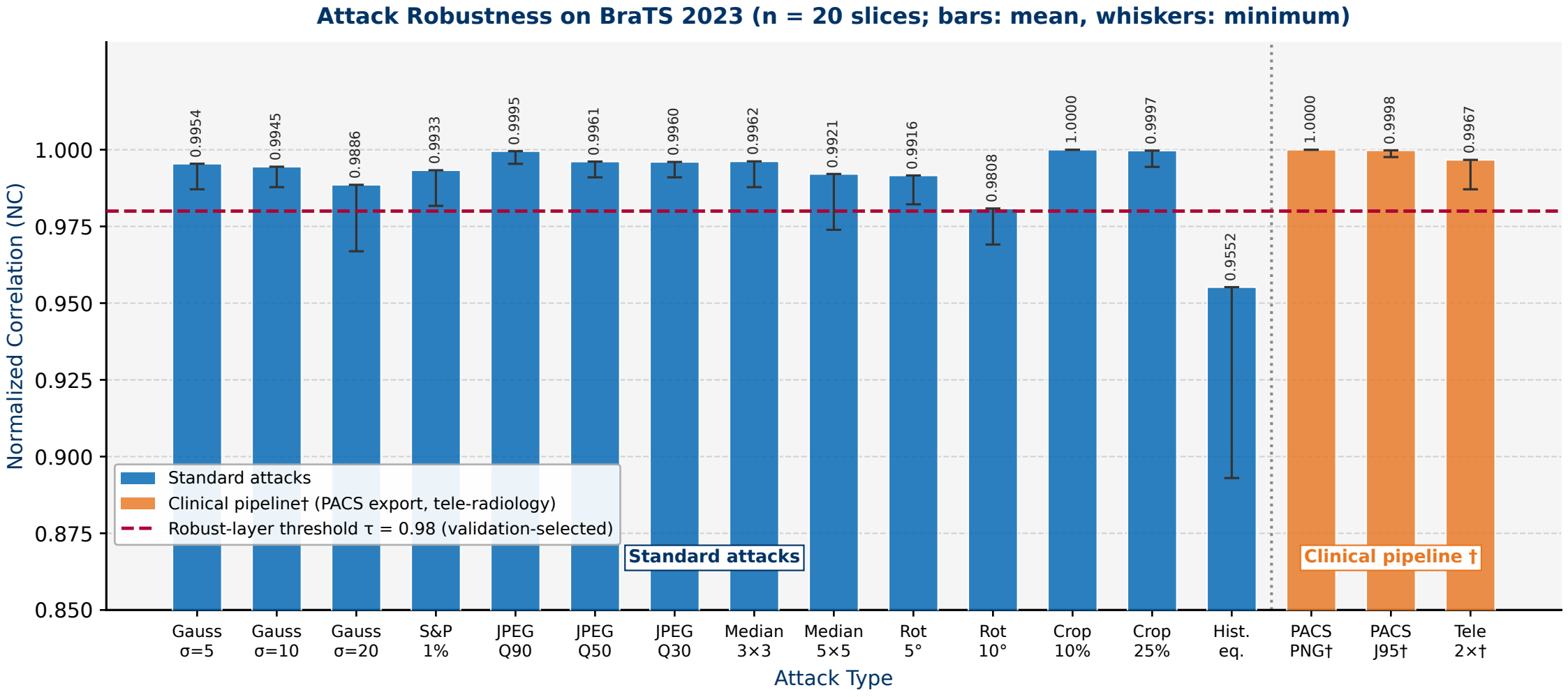
Attack Robustness on BraTS 2023 (n = 20 slices; bars: mean, whiskers: minimum)
Normalized Correlation (NC)
1.000
0.975
0.950
0.925
0.900
0.875
0.850
0.9954
0.9945
0.9886
0.9933
0.9995
0.9961
0.9960
0.9962
0.9921
0.9916
0.9808
1.0000
0.9997
0.9552
1.0000
0.9998
0.9967
Standard attacks
Clinical pipeline† (PACS export, tele-radiology)
Robust-layer threshold τ = 0.98 (validation-selected)
Standard attacks
Clinical pipeline †
Gauss σ=5
Gauss σ=10
Gauss σ=20
S&P 1%
JPEG Q90
JPEG Q50
JPEG Q30
Median 3×3
Median 5×5
Rot 5°
Rot 10°
Crop 10%
Crop 25%
Hist. eq.
PACS PNG†
PACS J95†
Tele 2×†
Attack Type